

Agentic Workflow Orchestrator

From Requirement to Production Release

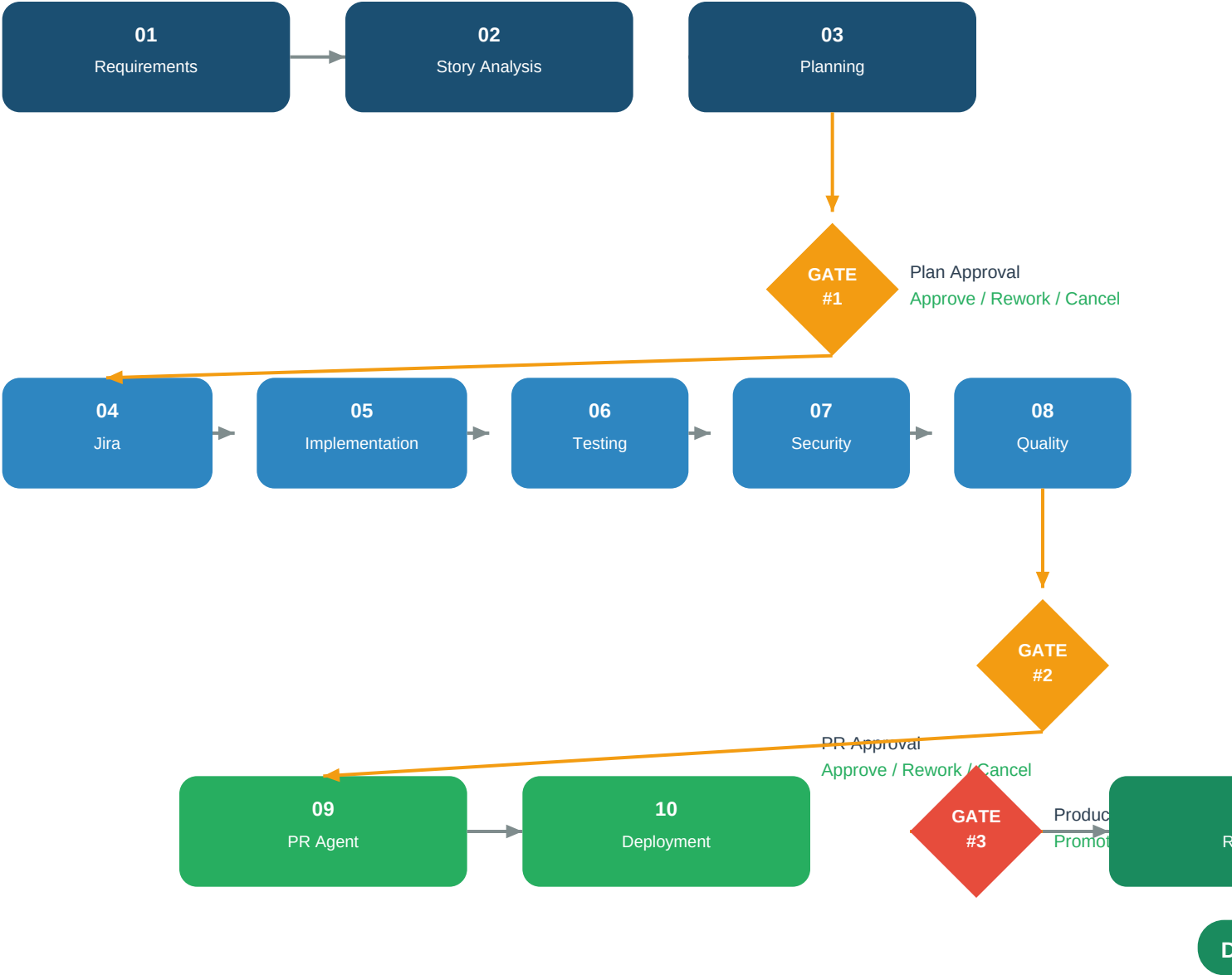
11 AI Agents | 3 Human Gates | Full SDLC

Amol Surjuse

Multi-Agent Architecture for Software Delivery

Agentic Workflow Orchestrator

The Full Agent Chain



Legend:



Agent

= AI Agent (automated)



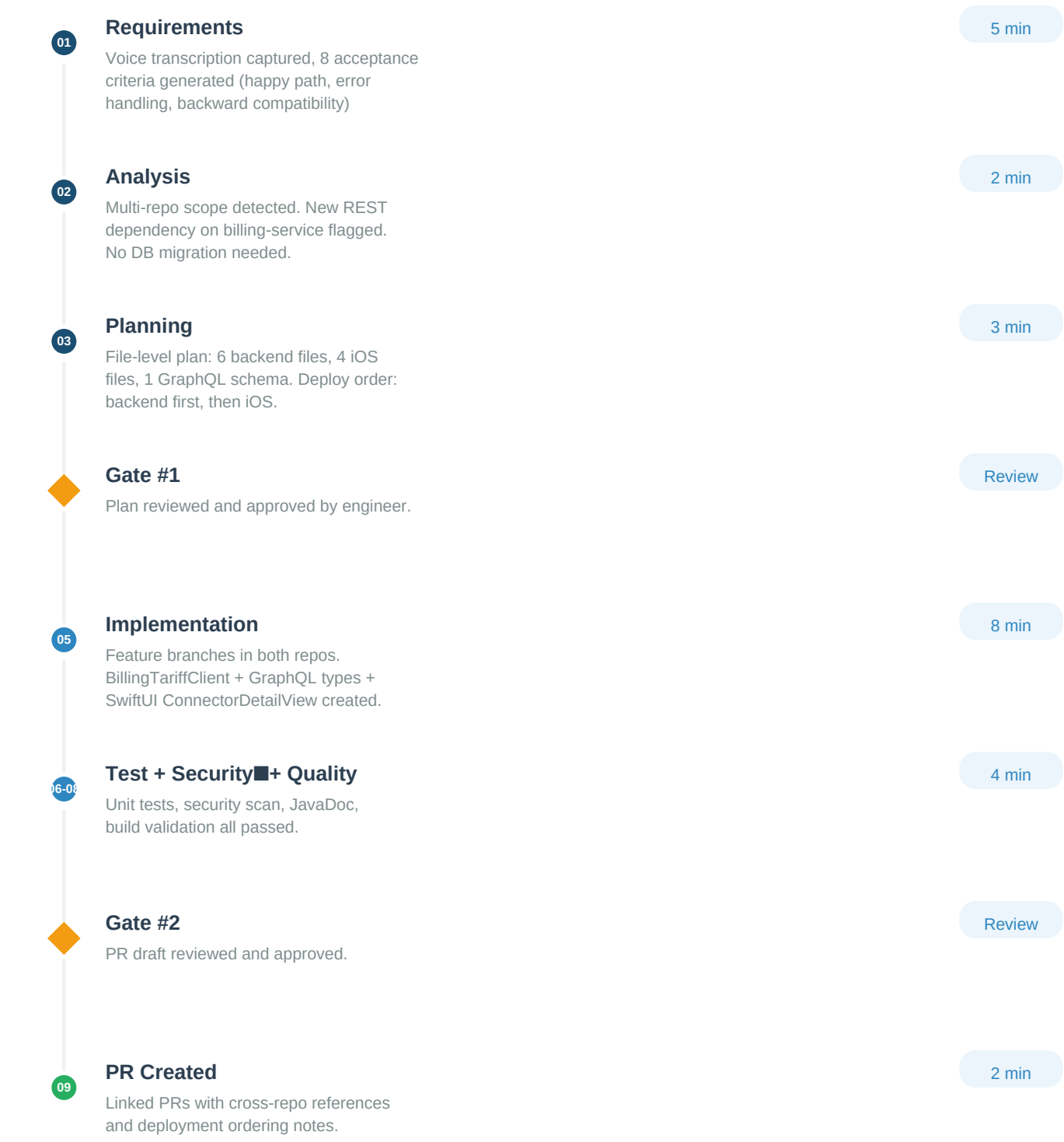
= Human Approval Gate (workflow pauses)

Agentic Workflow Orchestrator

Workflow 1: New Feature (Full Chain)

iOS Connector Pricing + Start Charging Button

Multi-repo: charger-management-service + driver-portal-ios + GraphQL schema

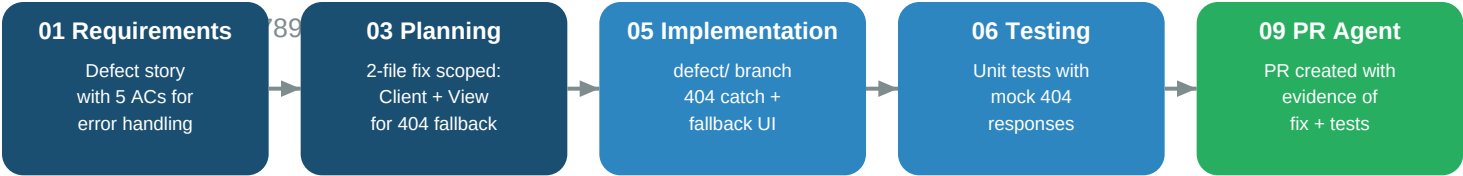


Result: Requirement to PR-ready in a single session

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Workflow 2: Bug Fix (Compressed Chain)

Connector Shows Available but Tariff Returns 404



Skipped: Agent 02 (deep analysis), Agent 07 (security), Agent 08 (quality)

Bug fixes go straight from requirements to planning — no deep analysis needed

What Changed:

`BillingTariffClient.java`

Catch 404 from billing-service, return `Optional.empty()`

`ConnectorDetailView.swift`

Show "Pricing unavailable" when tariffs list is empty

Acceptance Criteria (5 validated):

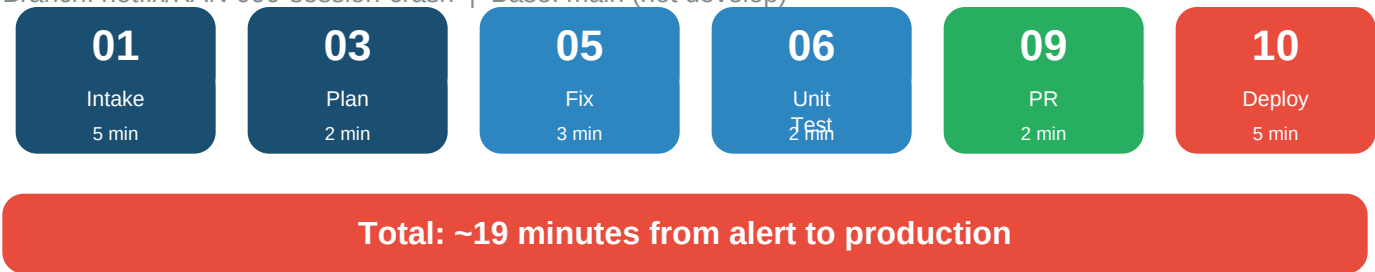
- AC-1: 404 response returns empty pricing (not crash)
- AC-2: UI shows fallback message for missing tariffs
- AC-3: Valid tariffs still display correctly alongside empty ones
- AC-4: Timeout from billing-service handled gracefully
- AC-5: Cache is not populated with failed lookups

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Workflow 3: Hotfix (Fast-Track)

PRODUCTION: Crash on Null TariffIds Array

Branch: hotfix/KAN-999-session-crash | Base: main (not develop)



How Hotfix Differs from Full Chain:

Skipped	Agent 02 (analysis), 07 (security), 08 (quality), 11 (release)
Abbreviated	Agent 01 does minimal intake, Agent 03 one-file plan
Testing	Unit test only — no integration, contract, or E2E
PR Target	main branch (not develop)
Deployment	Direct to production after smoke test — no staging hold
Gate #1	Still required — even hotfixes need plan approval

The Fix:

```
OcpChargerGraphQLService.java
List<String> tariffIds = connector.getTariffIds() != null
    ? connector.getTariffIds() : Collections.emptyList();
```

Every Story Requires Validated Acceptance Criteria

The Requirements Agent blocks handoff until all 4 validation checks pass



Count Check

Minimum ACs by story size: 3 (S), 5 (M), 8 (L/XL)



Coverage Check

Every functional requirement mapped to at least one AC



Negative Scenarios

At least one error/edge case scenario required



Format Check

Every AC follows Given/When/Then with verification method

Required Format:

AC-1: Available connector shows pricing details

Given a connector with status "Available" and tariffIds

When the user navigates to connector detail page

Then pricing displays energy rate, time rate, flat fees

Verification: Unit test + Integration test

Priority: MUST

ACCEPTANCE_CRITERIA_INCOMPLETE

If validation fails, Agent 01 blocks handoff and presents a gap report. No story proceeds without ACs.

Agentic Workflow Orchestrator

Key Design Principles

Single Responsibility

Each agent owns exactly one phase. The Requirements Agent doesn't deploy. The Deployment Agent doesn't write code. Narrow scope = reliable execution.

Standardized Contracts

Every agent communicates through Context Packets (inputs) and Output Packets (status, artifacts, exit criteria, next agent). Agents are independently testable and replaceable.

Human-in-the-Loop

Three approval gates at critical decision points: plan approval, PR approval, and production promotion. AI executes; the engineer decides.

Extensible Architecture

Adding a new agent = one SKILL.md file + one line in the orchestrator. No framework dependencies, no infrastructure. Just structured prompts with clear contracts.

Fail-Safe Design

20+ failure states with defined recovery paths. Every agent can report BLOCKED or REWORK_REQUIRED. The orchestrator never silently continues past a failure.

Runs entirely in VS Code through Claude Code

No framework | No infrastructure | Just structured prompts

11 Agents. 3 Gates. One Orchestrator.

From raw requirement to production release
with AI execution and human judgment.

What's next: scaling this pattern across teams
and making agent contracts a shared standard.

github.com/amolsurjuse

Amol Surjuse

#AgenticAI #SDLC #ClaudeCode #Automation